

Project 2: Market Risk & Volatility Dashboard

(using Monte Carlo Simulation)

Created By : Gauri Borse

Role : Data Analyst

➤ Project Title

Market Risk & Volatility Dashboard using Monte Carlo Simulation in Python

2. Objective

The objective of this project is to analyze stock market risk and price behavior using quantitative finance techniques. The system calculates daily returns, rolling volatility, Value at Risk (VaR), and performs Monte Carlo simulation to forecast possible future prices. An interactive dashboard is built using Streamlit to visualize results.

3. Problem Statement

Financial markets are uncertain and volatile. Investors and analysts need tools to measure risk, estimate volatility, and simulate possible future price movements. This project provides a data-driven risk analysis model and dashboard to support better financial decision-making.

4. Technologies Used

- Python
 - Pandas — data processing
 - NumPy — numerical simulation
 - Matplotlib — visualization
 - Streamlit — dashboard
 - CSV dataset — historical stock data
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5. Methodology / Working Steps

Step 1 — Data Collection

Historical stock price data is loaded from a CSV file using Pandas.

Step 2 — Daily Return Calculation

Percentage change of closing price is calculated to measure daily performance.

Step 3 — Rolling Volatility

A 20-day rolling standard deviation is computed to measure changing risk levels.

Step 4 — Monte Carlo Simulation

Multiple simulated future price paths are generated using:

- mean return
- volatility
- random normal distribution

Step 5 — Risk Metrics

The system calculates:

- Mean daily return
- Daily volatility

- Expected future price
- Value at Risk (95%)

Step 6 — Dashboard Visualization

A Streamlit dashboard displays:

- risk metrics
- returns graph
- volatility graph
- simulation forecast graph



Figure 1. Market Risk Dashboard Overview

6. Output Explanation

Daily Returns Graph

Shows day-to-day percentage change in stock price. Spikes indicate strong gain or loss days.

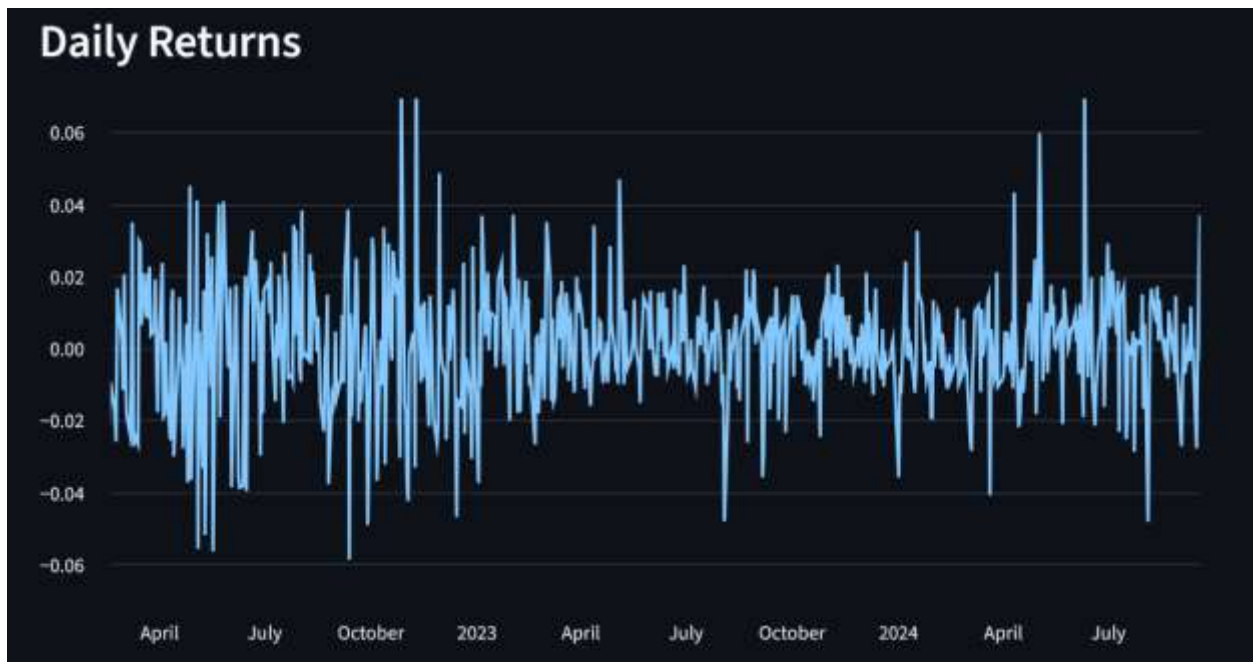


Figure 2. Daily Stock Returns Visualization

Rolling Volatility Graph

Shows how market risk changes over time using a 20-day moving window.



Figure 3. 20-Day Rolling Volatility Trend

Monte Carlo Simulation Graph

Displays hundreds of simulated future price paths. Each colored line represents one possible scenario of future movement.

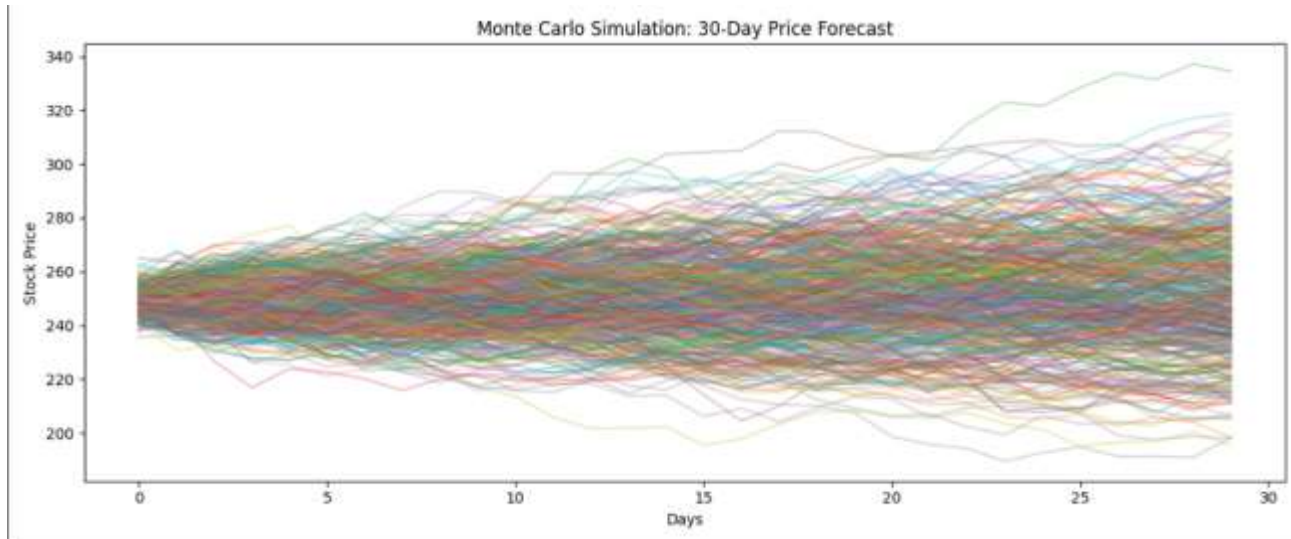


Figure 4. Monte Carlo Price Forecast Paths

Expected Price

Average predicted stock price after 30 simulated days.

Value at Risk (VaR 95%)

Indicates the maximum expected loss level at 95% confidence.

Expected Price (30 days)
253.43
Value at Risk (95%)
-0.02719

7. Real-World Applications

This type of model is used in:

- Investment banking
- Hedge funds
- Portfolio risk management
- Quantitative trading
- Financial analytics platforms
- FinTech risk engines

8. Conclusion

The project successfully implements quantitative risk modeling using Python. It demonstrates volatility measurement, probabilistic forecasting, and Monte Carlo simulation through an interactive dashboard. The system can support financial risk assessment and scenario analysis.